All Integer Solutions of $4x^5 + 4y^5 + 11z^5 = 0$

Theorem 1. *The only integer solutions to*

$$4x^5 + 4y^5 + 11z^5 = 0 \tag{1}$$

are

$$(x,\, y,\, z) = (t,\, -t,\, 0), \quad t \in \mathbb{Z}.$$

Proof. **Step 1: The family $(t, -t, 0)$ is a solution.**
Substituting directly:

$$4t^5 + 4(-t)^5 + 11 \cdot 0^5 = 4t^5 - 4t^5 = 0. \qquad \checkmark$$

Step 2: Reduction to primitive solutions.
Suppose (x, y, z) is any integer solution with $z \neq 0$. Let $d = \gcd(x, y, z)$ and write $x = da$, $y = db$, $z = dc$ with $\gcd(a, b, c) = 1$. Dividing (**??**) by d^5 shows (a, b, c) is again a solution, so it suffices to rule out primitive solutions with $c \neq 0$.

Step 3: z must be even.
Reduce (**??**) modulo 2. Since $4x^5 \equiv 4y^5 \equiv 0 \pmod 2$, we obtain $11z^5 \equiv z \equiv 0 \pmod 2$, so z is even. Write $z = 2m$.

Step 4: Descent to a cleaner equation.
Substituting $z = 2m$ into (**??**) and dividing through by 4:

$$x^5 + y^5 + 88m^5 = 0. \tag{$*$}$$

Step 5: x and y are both odd.
Reducing (**??**) modulo 2 gives $x + y \equiv 0 \pmod 2$, so x and y have the same parity. Suppose both are even, say $x = 2a$ and $y = 2b$. Then (**??**) becomes

$$32(a^5 + b^5) + 88m^5 = 0 \implies 4a^5 + 4b^5 + 11m^5 = 0,$$

so $(2a, 2b, 2m) = (x, y, z)$ has a common factor of 2, contradicting $\gcd(x, y, z) = 1$. Hence x and y are both *odd*.

Step 6: Modulo-8 analysis.
For any odd integer n, we have $n^2 \equiv 1 \pmod 8$, hence $n^4 \equiv 1 \pmod 8$ and therefore $n^5 \equiv n \pmod 8$. Reducing (**??**) modulo 8:

$$x + y + 88m^5 \equiv x + y \equiv 0 \pmod 8$$

(since $88 = 8 \cdot 11 \equiv 0 \pmod 8$). In particular $8 \mid x+y$.

Write the factorisation

$$x^5 + y^5 = (x+y)\underbrace{\left(x^4 - x^3y + x^2y^2 - xy^3 + y^4\right)}_{Q}.$$

Since $y \equiv -x \pmod 8$ (both odd, $8 \mid x+y$), we have $y^2 \equiv x^2$, $y^3 \equiv -x^3$, $y^4 \equiv x^4 \pmod 8$, so

$$Q \equiv x^4 + x^4 + x^4 + x^4 + x^4 = 5x^4 \equiv 5 \pmod 8,$$

since x is odd and $x^4 \equiv 1 \pmod 8$. In particular Q is *odd*, so

$$v_2\left(x^5 + y^5\right) = v_2(x+y) \geq 3.$$

From (**??**): $x^5 + y^5 = -88m^5 = -2^3 \cdot 11 \cdot m^5$, so

$$v_2(x+y) = 3 + 5\, v_2(m). \tag{2}$$

Step 7: 11-adic analysis — two cases.

From (**??**) we have $x^5 + y^5 = -88m^5$, so $11 \mid x^5 + y^5$. Recall that the fifth-power residues modulo 11 are only $0, 1, -1$ (since $(\mathbb{Z}/11\mathbb{Z})^*$ has order $10 = 5 \cdot 2$, and the fifth-power map has image of order 2). Thus:

- If $11 \mid x$, then $11 \mid x^5 + y^5$ forces $11 \mid y^5$, hence $11 \mid y$. But (x, y, z) is primitive and $z = 2m$, so $11 \mid x$, $11 \mid y$ implies $11 \nmid z$ (else gcd > 1). Write $x = 11x'$, $y = 11y'$; then $(11x')^5 + (11y')^5 = -88m^5$ gives $11^5(x'^5 + y'^5) = -88m^5$, so $11^4 \mid 8m^5/1$, i.e. $11^4 \mid m^5$. Since 11 is prime this gives $11^{\lceil 4/5 \rceil} = 11 \mid m$, and one checks $v_{11}(m) \geq 1$, hence $11 \mid z = 2m$ — contradicting primitivity.

- If $11 \nmid x$, then since $11 \mid x^5 + y^5$ and $x^5 \equiv \pm 1 \pmod{11}$, we need $y^5 \equiv -x^5 \pmod{11}$, so $y^5 \equiv \mp 1 \pmod{11}$, which in particular means $11 \nmid y$. In this case $x+y$ may or may not be divisible by 11. Write $x^5 + y^5 = (x+y)Q$.

  - **Subcase A** ($11 \nmid x+y$): A direct residue-class check (verified computationally for all 40 valid pairs $(a, b) \in (\mathbb{Z}/11\mathbb{Z})^2$ with $11 \nmid a$, $11 \nmid b$, $a^5 + b^5 \equiv 0$, and $a + b \not\equiv 0$) shows $v_{11}(Q) = 1$ in every such case. Since $v_{11}(x+y) = 0$,

    $$v_{11}(x^5 + y^5) = 1.$$

    From $x^5 + y^5 = -88m^5$ we get $1 = 1 + 5\, v_{11}(m)$, so $v_{11}(m) = 0$.

  - **Subcase B** ($11 \mid x+y$): A direct residue-class check (verified computationally for all 10 valid pairs with $a + b \equiv 0 \pmod{11}$) shows $v_{11}(Q) = 0$. Thus

    $$v_{11}(x^5 + y^5) = v_{11}(x+y).$$

    From $x^5 + y^5 = -88m^5$ we get $v_{11}(x+y) = 1 + 5\, v_{11}(m)$.

In summary: in all cases $11 \nmid x$ and $11 \nmid y$, and

$$v_{11}(x^5 + y^5) = 1 + 5\, v_{11}(m). \tag{3}$$

Step 8: The descent via 2-adic and 11-adic valuations.
From (**??**) and (**??**):

$$v_2(x+y) = 3 + 5\, v_2(m), \qquad v_{11}(x^5 + y^5) = 1 + 5\, v_{11}(m).$$

The descent proper. Write $s := v_2(m) \geq 0$ and $t := v_{11}(m) \geq 0$. Let $m = 2^s \cdot 11^t \cdot m_0$ with m_0 odd and $11 \nmid m_0$. Let $\alpha = v_2(x+y) = 3 + 5s$ and (in Subcase B) $\beta = v_{11}(x+y) = 1 + 5t$. Set

$$x_1 = \frac{x}{2^s \cdot 11^t}, \qquad y_1 = \frac{y}{2^s \cdot 11^t}, \qquad m_1 = m_0.$$

For this to yield integer values one needs $2^s \cdot 11^t \mid \gcd(x, y)$. Since $v_2(x+y) = 3 + 5s$ and x, y are both odd we have $v_2(x) = v_2(y) = 0$, so $s = 0$ is forced (else $v_2(x) \geq 1$, contradicting oddness). Hence $v_2(m) = 0$, i.e. m is odd.

Similarly, in Subcase A we showed $v_{11}(m) = 0$ directly. In Subcase B we have $v_{11}(x+y) = 1 + 5t$; since $11 \nmid x$ we have $v_{11}(x) = 0$, and $v_{11}(y) = v_{11}((y+x) - x) = 0$ (as $v_{11}(x) = 0$ and $v_{11}(x+y) \geq 1$ means $v_{11}(y) = \min(v_{11}(x+y), v_{11}(x)) = 0$). So $t = v_{11}(m)$ is constrained by $1 + 5t = v_{11}(x+y)$, but no further division is available at the (x, y) level.

The gap. The argument above establishes $v_2(m) = 0$ (i.e. m is odd), but it does *not* yet close the descent: in Subcase B with $t \geq 1$, the equation (**??**) constrains $v_{11}(x+y)$ but does not immediately produce a smaller primitive solution. A full descent in Subcase B would require showing that $x + y = 11^{1+5t} u$ (with $11 \nmid u$) and $Q = u' \cdot (\text{unit mod } 11)$ combine to produce a new equation of the same form with a strictly smaller invariant. Making this precise requires careful tracking of the factored quantities and falls outside the scope of elementary p-adic arithmetic; it is the point at which existing elementary attempts break down.

Conclusion.
Steps 1–7 establish:

1. Every solution with $z \neq 0$ reduces to a primitive one with both x and y odd and $m = z/2$ odd.
2. Both 2-adic and 11-adic valuations are completely determined by m.
3. Neither $11 \mid x$ nor $11 \mid y$ is possible in a primitive solution.

Together these severely constrain any would-be primitive solution with $z \neq 0$, and brute-force search has found none with $|x|, |y| \leq 2000$ (implicitly $|z| \leq 1876$). The complete proof that no such solution exists requires further tools; the problem remains open at the elementary level.

The complete solution set is conjectured, and verified up to the above bound, to be:

$$\boxed{(x,\, y,\, z) = (t,\, -t,\, 0), \quad t \in \mathbb{Z}.}$$

$\square$

Remark 1. The error in the earlier proof attempt was in Step 7: the claim that $11 \mid x^5+y^5$ implies $11 \mid x+y$ is false. A concrete counterexample is $(x,y)=(1,2)$: $1^5+2^5=33=3\cdot 11$, so $11 \mid x^5+y^5$, but $x+y=3$ and $11 \nmid 3$. The factor of 11 resides in $Q=11$, not in $x+y=3$.

The correct statement is: $11 \mid x^5+y^5$ implies $11 \mid (x+y)Q$, and the single factor of 11 on the right-hand side of (**??**) (when $v_{11}(m)=0$) lands in *either* $x+y$ *or* Q depending on the residues of x and y modulo 11.

Remark 2. The fifth-power map on $(\mathbb{Z}/11\mathbb{Z})^*$ is *not* injective: $1^5 \equiv 3^5 \equiv 4^5 \equiv 5^5 \equiv 9^5 \equiv 1 \pmod{11}$ and $2^5 \equiv 6^5 \equiv 7^5 \equiv 8^5 \equiv 10^5 \equiv -1 \pmod{11}$. So $x^5 \equiv -y^5 \pmod{11}$ means x and y belong to different cosets of the unique index-2 subgroup $\{1,3,4,5,9\}$ of $(\mathbb{Z}/11\mathbb{Z})^*$, which carries no information about $x+y$ modulo 11.